

Topic:-

Explain atomic models using materials such as old CDs, balloons, thread, marbles, etc.

Introduction:-

Everything in the universe around us—water, air, rocks, books, and even our own bodies—is made up of matter. Matter is composed of tiny fundamental particles called atoms. The word 'atom' comes from the Greek word '*Atomos*', which means indivisible. Ancient Indian philosopher Acharya Kanad had earlier stated that all matter is made of indestructible particles called *Parmanu*.

Because atoms are far too small to be seen even under the strongest optical microscope, scientists proposed theoretical models to explain what an atom looks like inside. Over the last two centuries, great scientists like John Dalton, J.J. Thomson, Ernest Rutherford, and Niels Bohr developed step-by-step atomic models. In this project, we used cheap, recycled craft items like old CDs, rubber balloons, marbles, and thread to build 3D physical models of these famous atomic structures.

❖ Three Fundamental Sub-Atomic Particles:-

- **Proton (p^+):** Positively charged particle inside the central nucleus.
- **Neutron (n^0):** Neutral particle (no charge) inside the nucleus with protons.
- **Electron (e^-):** Negatively charged tiny particle revolving around the nucleus.



Aim & Craft Materials Required:-

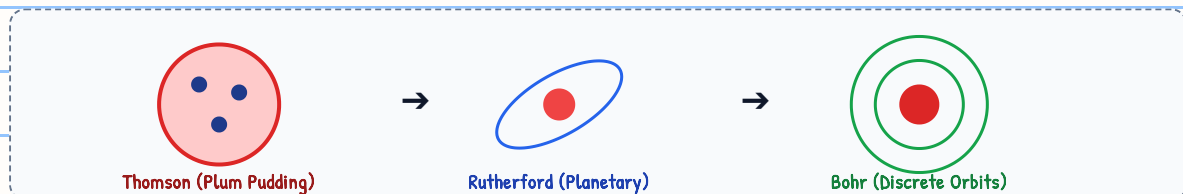
Aim: To design, construct, and present 3D working models of Thomson's, Rutherford's, and Bohr's atomic models using old CDs, balloons, thread, and marbles.

❖ *Objectives of the Project:-*

- To physically represent the historical evolution of atomic theory.
- To clearly differentiate between the nucleus and revolving electron orbits.
- To show why atoms are electrically neutral overall.

❖ *Eco-Friendly Materials Used:-*

- Old discarded CDs/DVDs for flat circular electron orbit rings.
- Rubber balloons, red clay balls, and glass marbles for protons and neutrons.
- Cotton thread, wooden sticks, black chart paper, and paper labels.



Procedure (How We Built the Models):-

We constructed the physical models following these creative steps:

- **1. Thomson's Plum Pudding Model (Watermelon):** We inflated a red round balloon halfway to represent the positive sphere of the atom. We glued small black paper dots (or black beads) all over the balloon to represent negative electrons embedded like seeds in watermelon pulp.
- **2. Rutherford's Nuclear Model:** We placed a dense group of marbles tightly glued in the center as the tiny, heavy nucleus. We tied long cotton threads around it, holding tiny buttons at the far ends to show that most of the atom is empty space.
- **3. Bohr's Orbit Model:** We took three concentric discarded CDs of increasing size mounted on a central wooden rod. The central spindle held red and yellow clay balls (protons and neutrons), while shiny beads were glued along the CD tracks to show electrons in K, L, M energy levels.

❖ Precautions During Model Assembly:-

- Handle scissors and CD edges carefully to prevent sharp cuts.
- Use eco-friendly paper glue instead of toxic adhesives.

Observations & Model Features Table:-

The table below compares the historical models represented by our craft items:

<i>Atomic Model</i>	<i>Scientist & Year</i>	<i>Craft Materials Used</i>	<i>Key Structural Feature</i>
Solid Sphere	John Dalton (1803)	Hard wooden / plastic ball	Hard, indivisible, solid sphere without charges
Plum Pudding	J.J. Thomson (1904)	Red balloon + black seeds	Positive sphere with embedded negative electrons
Nuclear Model	Ernest Rutherford (1911)	Central marbles + thread	Tiny central positive nucleus; mostly empty space
Planetary Orbit	Niels Bohr (1913)	Concentric CDs + beads	Electrons revolve in fixed circular energy shells (K, L, M)

Scientific Analysis & Discussion:-

Analyzing our physical models explains why each historical theory was modified:

- **Dalton's Limitation:** Dalton believed atoms were indivisible. Thomson disproved this by discovering the electron in 1897, proving that sub-atomic particles exist inside atoms!
- **Rutherford's Gold Foil Discovery:** Rutherford fired alpha particles at ultra-thin gold foil. Most passed straight through (proving an atom is 99.9% empty space!), but 1 in 20,000 bounced straight back, discovering the tiny, dense, positively charged nucleus.
- **Bohr's Stable Energy Shells:** Physics laws stated that revolving charges must radiate energy and spiral into the nucleus. Niels Bohr solved this puzzle by proving that electrons revolve only in discrete fixed orbits where they do not lose energy.

❖ Why Atoms Are Electrically Neutral:-

In every normal atom, the number of positive protons (+1) in the nucleus exactly equals the number of negative electrons (-1) in the shells, balancing total charge to zero.

Everyday Applications & Advantages:-

Understanding atomic models is the foundation for modern chemistry and technology:

- **Periodic Table of Elements:** Chemical bonding and properties of elements (metals, nonmetals) depend entirely on valence electrons in Bohr's outermost shell.
- **Electronics & Semiconductors:** Computers, mobile chips, and solar panels work by controlling electron flow across energy bands.
- **Nuclear Energy & Medicine:** Splitting uranium nuclei provides clean nuclear power, and radioactive isotopes cure cancer in radiation therapy.

❖ Advantages of Hands-on 3D Models:-

- Turns invisible microscopic physics into tangible, touchable educational art.
- Recycles discarded waste CDs and balloons instead of costly shop materials.
- Helps classmates easily remember electron capacity formulas: $2n^2$ (K=2, L=8, M=18).

What We Learned & Conclusion:-

- An atom consists of a tiny positive nucleus (protons + neutrons) surrounded by revolving electrons.
- Atomic theory evolved through experimental evidence from Dalton, Thomson, Rutherford to Bohr.
- Atoms are mostly empty space, yet their stable electronic shells form all matter in the universe.
- Building 3D models with recycled materials makes complex atomic physics simple and enjoyable.

Conclusion:-

Our project successfully built physical working models of Thomson's, Rutherford's, and Bohr's atomic structures using old CDs, balloons, thread, and marbles. By building these models with our own hands, our whole class visualized the inner architecture of the atom clearly, gaining a lasting appreciation for the brilliant scientists who unlocked the secrets of matter.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 5: Inside the Atom).
- NCERT Class 8 Science — Structure of the Atom.
- National Science Centre Hands-On Chemistry Learning Manual.